

The smallest semi-bimagic square of order six

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Abstract

A 6×6 array of distinct positive integers is *semi-bimagic* if all six rows and all six columns have the same sum S_1 and the same sum of squares S_2 (diagonals unconstrained). The oldest example, due to Pfeffermann (1894), uses integers up to 49. We show by exhaustive computer enumeration that the smallest possible largest entry is $\text{MaxNb} = 42$: semi-bimagic squares with distinct integers in $[1, 42]$ exist, and none exists with all entries in $[1, 41]$. At the optimum, $S_1 = 129$ is forced, and exactly four 36-element sets of integers admit such a square, with $S_2 \in \{3659, 3671, 3691, 3799\}$; all four sets are closed under $x \mapsto 43 - x$, a symmetry that emerges from the enumeration rather than being imposed. All exclusions 36–41 were confirmed by two independent algorithms with different underlying assumptions, and the classification at the optimum by two independent implementations jointly covering the full admissible window $S_1 \in [112, 147]$. As a corollary, squaring the minimal square cell-wise yields a 6×6 semi-magic square of 36 distinct perfect squares with constant 3659.

A second, structural part isolates the local mechanism by which two columns of a semi-bimagic square interact: *twin triples*, three-element subsets of the two column sets with equal sums and equal sums of squares (degree-2 Prouhet–Tarry–Escott pairs). Twins aligned at three shared positions under the row pairing are exactly the perturbations of the square confined to two columns; twins aligned at two are the skew exchanges. Empirically the two realizations exclude each other for every tested pairing of every column-block pair arising from the fifteen squares surveyed, across more than 420,000 explicit pairings. We prove (Theorem 13) that the exclusion holds for *every* pairing whenever the pair of sextuples admits at most one twin up to complementation — which holds, with no exception, for all 1,078 column-block pairs arising from the fifteen squares surveyed. An exhaustive census of *free* bimagic pairs over $[1, N]$, $N \leq 26$, then identifies, within the census range, the two residual forms at which the general conjectures fail: pairs with four twins occur, in the two incidence forms singled out by our case analysis; their alignment statistics depend only on the form, as reproduced by an abstract six-point model, and one form admits mixed pairings whereas the other has none. Neither form occurs among the column blocks of the fifteen squares surveyed. The uniqueness condition holds throughout the surveyed square environment and is conjectured as an environmental property.

1 Introduction

A magic square is *bimagic* if it remains magic after squaring each entry, and *semi-bimagic* if rows and columns (but not necessarily the diagonals) have constant sum and constant sum of squares. The first semi-bimagic square of order six with distinct integers was published by G. Pfeffermann in 1894 [1]; it uses integers up to 49, with $S_1 = 150$ and $S_2 = 5150$, and has

remained the reference artifact for 132 years. The systematic record of this field is maintained by Christian Boyer [2].

For *fully* bimagic squares of order six with distinct integers, the minimal known example is due to Lee Morgenstern (2006): $(\text{MaxNb}, S_1, S_2) = (72, 219, 10663)$, proved minimal in 2008–09 [2]. To our knowledge, the corresponding minimality question for the *semi*-bimagic case — how small can the largest entry be when the diagonal constraints are dropped — has not been recorded, either as a problem or as a record [2].

Our contribution is the answer at order six: $\text{MaxNb} = 42$, with the full classification of the optimal sets. The pair of results (Morgenstern’s $(72, 219, 10663)$ versus our $(42, 129, 3659)$) also quantifies, for the first time at order six, the cost of the two diagonal constraints.

The second half of the paper (Section 7) turns from the record to the mechanism. The enumeration shows that survival is decided almost entirely at the orthogonality stage, and Remark 20 that no simple counting invariant governs it. We isolate the local objects that do carry structure: *twin triples*, degree-2 Prouhet–Tarry–Escott pairs between two disjoint bimagic sextuples, whose two possible realizations under a pairing (row-aligned and skew) we prove to correspond bijectively to the homogeneous and affine local exchanges of a square (Proposition 9). On every pair arising from the fifteen squares surveyed, the two realizations exclude each other. Theorem 13 proves this exclusion for every pairing whenever the pair carries at most one twin up to complementation, and a screen of all 1,078 column-block pairs across those squares finds no exception to that uniqueness. On free pairs of bimagic sextuples the picture is sharper still: an exhaustive census up to $[1, 26]$ (Section 7.5) shows that four-twin pairs occur, in exactly the two configurations left open by the case analysis, and that within the census the general exclusion fails on one of them and holds on the other — uniqueness is thus conjectured as a property of the *square environment* (Conjecture 16).

Exhaustive enumeration is the accepted standard of proof in this area: the impossibility of a 7×7 bimagic square of consecutive integers is accepted on the basis of four independent programs, and Morgenstern’s minimality on months of dedicated computation [2]. We meet that bar by running two algorithms with different assumptions on every exclusion (Section 4).

2 Definitions and statements

Definition 1. A semi-bimagic square of order 6 is a 6×6 array of pairwise distinct positive integers such that all rows and all columns have the same sum S_1 and the same sum of squares S_2 . Its MaxNb is its largest entry.

Theorem 2 (Minimality). *Semi-bimagic squares of order 6 with $\text{MaxNb} = 42$ exist, and no semi-bimagic square of order 6 has $\text{MaxNb} \leq 41$.*

Theorem 3 (Classification at the optimum). *Every semi-bimagic square of order 6 with $\text{MaxNb} = 42$ has $S_1 = 129$, and its set of 36 entries is one of exactly four sets, namely $[1, 42]$ minus one of*

$$\{19, \dots, 24\}, \quad \{4, 11, 18, 25, 32, 39\}, \quad \{7, 8, 21, 22, 35, 36\}, \quad \{10, 11, 12, 31, 32, 33\},$$

with $S_2 = 3799, 3659, 3671, 3691$ respectively. All four sets are closed under $x \mapsto 43 - x$.

Remark 4 (Independent replication). *Theorem 3 rests on two independent implementations. The second (a different framing, written independently) enumerated all candidate sets over the full admissible window $S_1 \in [112, 147]$ without assuming 1 in the set, examining every candidate admitting at least one row partition (45,985 entries in the published data). Its success condition was pre-registered before the run returned: exactly four sets with a square, all at $S_1 = 129$, exactly the four sets above — a square found anywhere else would have falsified the theorem. The outcome: the four sets above, with row partition counts 132, 50, 83, 59 matching the first implementation, and no square elsewhere. A pre-registered negative control behaved as required: the centred set omitting $\{2, 3, 4, 39, 40, 41\}$ (with $S_2 = 3459$, below the realized minimum) admits six row partitions and no square. As a further internal cross-check, the per-slot totals of examined candidates (84,948; 76,605; 76,605; 84,948 over the four slots, 323,106 in all) satisfy an exact mirror identity: the map $x \mapsto 42 - x$ on the omitted 6-subsets of $\{1, \dots, 41\}$ (the entry 42 belongs to every candidate set) sends stratum S_1 to stratum $259 - S_1$ and preserves the candidacy conditions, pairing the slots as (112–126, 133–147) and (127–129, 130–132) — the examined counts pair exactly, while the partitionable counts rightly do not, since no entrywise bijection transports the semi-bimagic structure.*

3 The squares

The minimal- S_2 square, (42, 129, 3659), on the set omitting $\{4, 11, 18, 25, 32, 39\}$ (an arithmetic progression of step 7):

$$\begin{pmatrix} 1 & 12 & 20 & 27 & 33 & 36 \\ 28 & 31 & 37 & 2 & 10 & 21 \\ 38 & 23 & 35 & 14 & 16 & 3 \\ 13 & 15 & 5 & 40 & 22 & 34 \\ 19 & 6 & 8 & 29 & 41 & 26 \\ 30 & 42 & 24 & 17 & 7 & 9 \end{pmatrix}$$

A square on the set omitting the six central consecutive integers $\{19, \dots, 24\}$, (42, 129, 3799):

$$\begin{pmatrix} 1 & 6 & 26 & 29 & 33 & 34 \\ 27 & 28 & 31 & 36 & 2 & 5 \\ 32 & 35 & 3 & 4 & 25 & 30 \\ 40 & 39 & 17 & 14 & 12 & 7 \\ 11 & 8 & 42 & 37 & 16 & 15 \\ 18 & 13 & 10 & 9 & 41 & 38 \end{pmatrix}$$

A 26/28 square (two magic, non-bimagic diagonals, matching Pfeffermann’s diagonal pattern but with $\text{MaxNb} = 42$):

$$\begin{pmatrix} 1 & 26 & 29 & 34 & 33 & 6 \\ 27 & 31 & 36 & 5 & 2 & 28 \\ 32 & 3 & 4 & 30 & 25 & 35 \\ 18 & 10 & 9 & 38 & 41 & 13 \\ 11 & 42 & 37 & 15 & 16 & 8 \\ 40 & 17 & 14 & 7 & 12 & 39 \end{pmatrix}$$

All three were re-verified by an independent checker calibrated on the four historical squares (see Data availability).

4 Method: two independent algorithms

Algorithm 1 (set-sweep). Enumerates candidate 36-element sets, then partitions into six bimagic sextets (equal S_1 and S_2), then column extensions with pruning. It does *not* assume $1 \in \text{set}$: no translation lemma is used.

Algorithm 2 (bucket). Enumerates all sextets of $[1, M]$, buckets them by (S_1, S_2) , and searches for six pairwise disjoint sextets within a bucket; it assumes $1 \in \text{set}$ and $M \in \text{set}$ via the translation lemma.

The two implementations therefore differ in their hypotheses, not only in their code. Every exclusion below carries both.

MaxNb	outcome	algorithms	logs
36	excluded	set-sweep + bucket + brute-force sextets	blim3yxim, groups36.log, verify36.log
37	excluded	set-sweep + bucket	blim3yxim, bn0vyvt60
38	excluded	set-sweep + bucket	blim3yxim, bn0vyvt60
39	excluded	set-sweep + bucket	b62iegrr5, groups39.log
40	excluded	bucket + set-sweep (82,251 sets)	bjzvtqu5c, bz9px7zym
41	excluded	bucket + set-sweep (658,008 sets)	bjzvtqu5c, bz9px7zym
42	attained + classified	two independent implementations, full window (Remark 4)	bjzvtqu5c, b5zmv01yv, b7x4hoknj, bmlofdsyf, replica-full-s1_*.json, replica-full-verdetto.log

5 The four optimal sets and the collapse

At $\text{MaxNb} = 42$, $S_1 = 129$ is forced. The exhaustive phase-2 run found 324 partitions of candidate sets into six bimagic rows, of which exactly 24 extend to a full square, distributed over the four sets:

omitted set	S_2	partitions (extending/total)	structure
$\{19, \dots, 24\}$	3799	8 / 59	six central consecutive
$\{4, 11, 18, 25, 32, 39\}$	3659	8 / 132	arithmetic progression, step 7
$\{7, 8, 21, 22, 35, 36\}$	3671	4 / 50	three adjacent pairs
$\{10, 11, 12, 31, 32, 33\}$	3691	4 / 83	two blocks of three

Both totals (324 and 24) were replicated by an independent implementation with a different framing.

Where candidates die. Within the symmetric class, the counting chain has four declarable stages:

$$546 \text{ arithmetic candidates} \xrightarrow{\text{sextet cover}} 546 \xrightarrow{\text{row partitions}} 382 \xrightarrow{\text{orthogonality}} 4.$$

No candidate dies at the sextet-cover stage; 164 die for lack of any row partition; and 378 of the remaining 382 (99.0%) die at the column-extension (orthogonality) stage. The bottleneck is orthogonality, not arithmetic. A striking witness: the centred set omitting $\{2, 3, 4, 39, 40, 41\}$ passes all arithmetic conditions with $S_2 = 3459$ — *below* the realized minimum — and admits six row partitions, yet no square exists on it.

Symmetry as output. All four optimal sets are closed under $x \mapsto 43 - x$. This is not a filter of the search: $S_1 = 129$ only forces the mean 21.5, and the enumeration is symmetry-agnostic. The central symmetry emerges.

A finer instance. The omitted set of the S_2 -minimal winner is exactly the residue class 4 (mod 7) of $\{1, \dots, 42\}$. The mirror $x \mapsto 43 - x$ sends class r to class $1 - r \pmod{7}$, and $r = 4$ is the *unique* self-mirror class. Its five sister classes are admissible candidates in their own right ($S_1(r) = 133 - r$, $S_2(r) = 3815 - 35r - r^2$, always integral), yet none of them admits a single row partition: among the six whole-class omissions, the unique symmetric one is the only one to reach the partition stage — and it wins.

Proposition 5 (Lifting of the mirror symmetry). *The involution $\sigma : x \mapsto 43 - x$ sends bimagic sextets of a mirror-closed entry set to bimagic sextets ($\sum(43 - x) = 258 - 129 = 129$; $\sum(43 - x)^2 = 11094 - 2 \cdot 43 \cdot 129 + \sum x^2 = \sum x^2$), hence acts on the row partitions of each optimal set and on orthogonal pairs of partitions, preserving the intersection profile. Orbits have size one or two, so an odd count of orthogonal pairs forces a σ -fixed pair, whose square satisfies $43 - M = M$ up to row and column permutations (self-complementary); a non-fixed pair is exchanged with its σ -image, the two squares being complements of one another up to permutations, composed with transposition when σ swaps the row and column roles. With the orthogonal-pair counts 5, 5, 2, 2: the $S_2 = 3799$ set carries one self-complementary square and two plainly complement-paired orbits; 3659 one self-complementary square, one plain and one transposed orbit; 3671 one plain orbit; 3691 one transposed orbit (all verified explicitly). The mirror symmetry of the optimal sets thus lifts to the optimal squares themselves.*

6 A corollary on squares of squares

Squaring the $(42, 129, 3659)$ square cell-wise gives a 6×6 *semi-magic square of 36 distinct perfect squares* with constant 3659 and largest entry $42^2 = 1764$. (Magic squares of squares of order 6 with diagonals are known: Boyer 2005 [3]; the point here is only the size of the entries, inherited from the record.)

7 The local structure of orthogonality: twin triples

The enumeration of Section 5 shows that survival is decided almost entirely at the orthogonality stage (99.0% of the row-partitionable symmetric candidates die there), and Remark 20 that neither symmetry nor partition abundance governs that stage. This section opens the

structural study of the mechanism itself. The natural local objects are pairs of columns: two columns of a semi-bimagic square are two disjoint six-element sets with the same sum and the same sum of squares, interacting through the rows. We first develop the combinatorics of such pairs in isolation, then return to the square.

7.1 Twins, alignment, realizations

Definition 6 (bimagic pair, twins). A bimagic pair is a pair (A, B) of disjoint six-element sets of positive integers with $\sum A = \sum B$ and $\sum_{a \in A} a^2 = \sum_{b \in B} b^2$. A twin of (A, B) is a pair $g = (T_A, T_B)$ of three-element subsets $T_A \subset A$, $T_B \subset B$ with

$$\sum T_A = \sum T_B \quad \text{and} \quad \sum_{x \in T_A} x^2 = \sum_{y \in T_B} y^2,$$

i.e. an instance of the Prouhet–Tarry–Escott problem of degree 2 and size 3 [4] inside the pair. We write $G(A, B)$ for the set of twins of (A, B) , and $\bar{g} = (A \setminus T_A, B \setminus T_B)$ for the complement of a twin.

Complementation is well defined: $\sum(A \setminus T_A) = \sum A - \sum T_A = \sum B - \sum T_B = \sum(B \setminus T_B)$, and likewise for squares. Since $T_A \neq A \setminus T_A$, the map $g \mapsto \bar{g}$ is a fixed-point-free involution of $G(A, B)$; in particular $|G(A, B)|$ is always even.

Lemma 7 (rigidity). Two three-element subsets of the same six-element set sharing two elements and having the same sum coincide. Consequently, if two distinct twins share exactly two elements on some side, their invariants $(\sum T, \sum T^{(2)})$ differ.

Proof. The third element is determined already by the linear sum. $\square$

Definition 8 (pairing, alignment). A pairing of a bimagic pair is a bijection $\pi: A \rightarrow B$; when A, B are two columns of a square, the rows provide one. The alignment of a twin $g = (T_A, T_B)$ under π is

$$k(g, \pi) = |\pi(T_A) \cap T_B| \in \{0, 1, 2, 3\},$$

and we write $N_3(\pi)$, $N_2(\pi)$ for the numbers of twins of (A, B) with $k = 3$, resp. $k = 2$.

Twins aligned at 3 and at 2 are precisely the two ways a twin can be *realized* on a square. Fix two columns c_1, c_2 of a semi-bimagic square M ; its entries being pairwise distinct, so are in particular the twelve paired entries $p_j = M_{j,c_1}$, $q_j = M_{j,c_2}$ (j the row) — the hypothesis used below to exclude $|J| \in \{1, 2\}$ and to recover rows uniquely. Write $d_j = q_j - p_j$, $e_j = q_j^2 - p_j^2$; the row pairing is $\pi(p_j) = q_j$.

Proposition 9 (realizations). (i) (row-aligned twins are the moves confined to two columns.) For a set of rows J with $0 < |J| < 6$, swapping the entries p_j and q_j in the rows $j \in J$ yields again a semi-bimagic square if and only if $\sum_{j \in J} d_j = 0$ and $\sum_{j \in J} e_j = 0$ (homogeneous system). Every such J has $|J| = 3$, and

$$J \longmapsto (\{p_j: j \in J\}, \{q_j: j \in J\})$$

is a bijection from the proper solutions of the homogeneous system onto the twins with $k(g, \pi) = 3$.

(ii) (skew twins.) *Call affine system the set of triples (j_1, j_2, K) with $j_1 \neq j_2$, K a two-element subset of the remaining four rows, and*

$$q_{j_1} - p_{j_2} + \sum_{j \in K} d_j = 0, \quad q_{j_1}^2 - p_{j_2}^2 + \sum_{j \in K} e_j = 0.$$

Then $(j_1, j_2, K) \mapsto (\{p_{j_2}\} \cup \{p_j : j \in K\}, \{q_{j_1}\} \cup \{q_j : j \in K\})$ is a bijection from the affine system onto the twins with $k(g, \pi) = 2$.

Proof. (i) Each row keeps its two entries under the swap, so all row sums and row square-sums are unchanged, as are the other columns; column c_1 changes by $\sum_J d_j$ in sum and $\sum_J e_j$ in square-sum (and c_2 by the negatives), whence the criterion. Sizes: the full set $J = \{1, \dots, 6\}$ solves the system trivially (the two columns have equal sums and equal square-sums), so complements of solutions are solutions; $|J| = 1$ forces $d_j = e_j = 0$, i.e. $p_j = q_j$, excluded; if $J = \{i, j\}$ has size 2, then $d_i = -d_j$, and $e = d(p + q)$ turns $\sum e = 0$ into $p_i + q_i = p_j + q_j$; combined with $d_i = -d_j$ this gives $q_i = p_j$ and $p_i = q_j$, repeated entries — excluded. So proper solutions have $|J| = 3$. The map lands on twins: the sums of $\{p_j\}_J$ and $\{q_j\}_J$ agree iff $\sum_J d = 0$, likewise for squares; and $\pi(\{p_j\}_J) = \{q_j\}_J$ gives $k = 3$. It is injective (J is the set of rows of T_A), and surjective: if $k(g, \pi) = 3$ then $\pi(T_A) = T_B$, so with J the rows of T_A we get $\sum_J d = \sum T_B - \sum T_A = 0$, and likewise for e .

(ii) The two displayed equations say exactly that the two triples have equal sums and equal square-sums. Alignment: $\pi(T_A) = \{q_{j_2}\} \cup \{q_j\}_{j \in K}$, and since $j_1 \neq j_2$ and $j_1, j_2 \notin K$, the intersection with $T_B = \{q_{j_1}\} \cup \{q_j\}_{j \in K}$ is $\{q_j\}_{j \in K}$, so $k = 2$. Conversely, from a twin with $k = 2$ one recovers uniquely: $K =$ the rows j with $p_j \in T_A$ and $q_j \in T_B$ (exactly two of them), $j_2 =$ the row of the element of T_A not paired into T_B , $j_1 =$ the row of the element of T_B not hit by $\pi(T_A)$; and $j_1 \neq j_2$, since $j_1 = j_2$ would give $k = 3$. $\square$

Remark 10 (machine check). *Both bijections were verified computationally by complete enumeration of both sides on 32,400 pairings (45 column pairs from three sample squares of different families $\times$ all 720 bijections): 1,008 homogeneous solutions against 1,008 twins at $k = 3$ and 9,072 affine solutions against 9,072 twins at $k = 2$, with multiset equality and injectivity holding on every single pairing.*

7.2 The exclusion phenomenon

On the actual squares the two realizations never coexist. On all 1,200 column pairs of the seven squares in our test harness (two of the optimal squares, an auxiliary square with entries up to 44 produced during our exploratory searches, Pfeffermann's and Boyer's squares, Wroblewski's square with row sum 648, and Morgenstern's with row sum 330), the homogeneous system is solvable on 64 pairs — every solution with $|J| = 3$, as Proposition 9 forces — and *no pair solves both systems*. The real row pairing is not what protects: permuting the pairing exhaustively — a pre-registered deterministic shuffle, 3,600 pairings — homogeneous solutions do appear where the real pairing has none, yet still no pairing solves both. Neither does any other pairing: on 45 column pairs the full sweep of all 720 bijections each (32,400 pairings) finds $N_3(\pi)N_2(\pi) = 0$ throughout, and a wider screen (388,800 pairings: the $S_2 = 3799$ optimal square, Pfeffermann's, and Wroblewski's row-sum-648 square — three different structural families — over all 36 oriented orthogonal views, all 15 column pairs and all 720 bijections) finds the same. This motivates the general conjecture:

Conjecture 11 (exclusion, general form). *For every bimagic pair (A, B) and every pairing π ,*

$$N_3(\pi) \cdot N_2(\pi) = 0.$$

Section 7.5 refutes Conjecture 11 in this generality. The theorem below identifies the hypothesis under which its conclusion does hold, for every pairing.

7.3 Exclusion from uniqueness: Theorem G

Lemma 12 (complementation preserves alignment). *For every twin g and pairing π : $k(\bar{g}, \pi) = k(g, \pi)$.*

Proof. $\pi(A \setminus T_A) = B \setminus \pi(T_A)$, so

$$|\pi(A \setminus T_A) \cap (B \setminus T_B)| = 6 - |\pi(T_A) \cup T_B| = 6 - (3 + 3 - k(g, \pi)) = k(g, \pi). \quad \square$$

Theorem 13 (exclusion from uniqueness). *Let (A, B) be a bimagic pair with $G(A, B) = \emptyset$ or $G(A, B) = \{g, \bar{g}\}$ (at most one twin up to complementation). Then $N_3(\pi) N_2(\pi) = 0$ for every pairing π .*

Proof. If $G = \emptyset$ there is nothing to align. Let $G = \{g, \bar{g}\}$. Alignment under a fixed π assigns to each twin a single value k , and by Lemma 12 the two twins carry the same value. If $N_3(\pi) > 0$, some twin has $k = 3$, hence both do; G is exhausted by twins aligned at 3, and $N_2(\pi) = 0$. $\square$

The proof is pure combinatorics of finite sets: the integer values enter only through the definition of G . Theorem 13 concentrates all the arithmetic of Conjecture 11 into a statement about twelve integers:

Conjecture 14 (uniqueness of twins, general form). *Every bimagic pair admits at most one twin up to complementation: $G(A, B) = \emptyset$ or $G(A, B) = \{g, \bar{g}\}$.*

Corollary 15. *The uniqueness condition implies the exclusion property: Conjecture 14 implies Conjecture 11.*

Conjecture 14 too is refuted in general by the free census (Section 7.5): four-twin pairs exist. What the fifteen-square data support exhaustively — and what we conjecture — is its environment form:

Conjecture 16 (uniqueness in the square environment). *For every pair (A, B) of column blocks of the partitions entering a semi-bimagic square of order 6: $G(A, B) = \emptyset$ or $G(A, B) = \{g, \bar{g}\}$.*

By Theorem 13, Conjecture 16 yields the exclusion on every pairing throughout the square environment.

7.4 Evidence, and the residual case

Direct census. On the 45 column pairs of the three sample views of Remark 10: $|G| \in \{0, 2\}$ with no other value (31 empty, 14 non-empty), and every non-empty G is a complementary pair $\{g, \bar{g}\}$. On the wide base: among all 1,078 distinct pairs of column blocks taken from the partitions entering orthogonal pairs across fifteen squares (the four optimal squares, the auxiliary square with entries up to 44, and ten historical or recorded squares: Pfeffermann's, Boyer's, two of Morgenstern's, four of Wroblewski's and two of Rouanet's):

$$|G| = 0 \text{ on 811 pairs,} \quad |G| = 2 \text{ on 267 pairs,}$$

always a complementary pair, with no violation. The status of this evidence is declared: exhaustive in the sample, not a proof.

Remark 17 (the sweep is universal). *For a pair with $G = \{g, \bar{g}\}$ the distribution of alignments over all 720 pairings is forced by pure combinatorics, independently of the values: the number of bijections of two six-element sets carrying a fixed 3-subset onto a fixed 3-subset with intersection exactly k is 36, 324, 324, 36 for $k = 3, 2, 1, 0$. Hence every non-empty pair shows the same sweep — 36 pairings with $N_3 > 0$, 324 with $N_2 > 0$, 360 with neither — exactly as observed on all 14 non-empty pairs of the census. The same four numbers reappear as the row marginals of the alignment tables of every four-twin pair of the free census (Section 7.5).*

The residual case. What remains between the evidence and Conjecture 14 is the exclusion of two *independent* twins — two twins that are neither equal nor complementary. The route through complements, suggested to us in review, closes most of the possible configurations:

Proposition 18 (reduction of the residual case). *Let $g = (T_A, T_B)$ and $g' = (T'_A, T'_B)$ be independent twins of a bimagic pair (A, B) — twins that are neither equal nor complementary, $g' \notin \{g, \bar{g}\}$ — and $(a, b) = (|T_A \cap T'_A|, |T_B \cap T'_B|)$. Up to replacing g' by $\bar{g}'$ (which turns (a, b) into $(3 - a, 3 - b)$) and swapping the two sides, only two configurations are possible: (I) half-sum, $(a, b) = (3, 0)$: both triples of g have halved invariants, $2\sum T = \sum A$ and $2\sum T^{(2)} = \sum A^{(2)}$, and then $|G(A, B)| \geq 4$; (II) transversal, $(a, b) = (2, 1)$. Consequently Conjecture 14 holds if and only if no bimagic pair carries a twin with halved invariants or a transversal pair of twins.*

Proof. The cells $(3, 3)$ and $(0, 0)$ are $g' = g$ and $g' = \bar{g}$, excluded by independence. Let $a = 3$, i.e. $T'_A = T_A$ (this covers $a = 0$ by complementation and $b \in \{0, 3\}$ by the side swap): then T'_B and T_B have the same invariants, both equal to those of T_A . Now $b = 2$ is impossible by Lemma 7. For $b = 1$, write $T_B = \{x, u, v\}$ and $T'_B = \{x, r, s\}$ with x the shared element: equal sums give $u + v = r + s$, equal square-sums give $u^2 + v^2 = r^2 + s^2$, hence $2uv = (u + v)^2 - (u^2 + v^2) = 2rs$; so $\{u, v\}$ and $\{r, s\}$ are the root pairs of the same quadratic $X^2 - (u + v)X + uv$, i.e. $\{u, v\} = \{r, s\}$ and $T'_B = T_B$ — contradicting $b = 1$. Finally $b = 3$ gives $g' = g$. So $b = 0$: $T'_B = B \setminus T_B$, whose invariants equal those of T_B exactly when both are halved — configuration (I) — and then $g, \bar{g}, (T_A, B \setminus T_B), (A \setminus T_A, T_B)$ are four distinct twins. Let $a, b \in \{1, 2\}$: for $(a, b) = (2, 2)$, subtracting the twin conditions of g' from those of g the shared parts cancel, leaving $a_3 - a_4 = b_3 - b_4$ and $a_3^2 - a_4^2 = b_3^2 - b_4^2$ for the third elements, with $a_3 \neq a_4$; hence $a_3 + a_4 = b_3 + b_4$, so $a_3 = b_3$ — impossible, as $A \cap B = \emptyset$.

The cell $(1, 1)$ reduces to $(2, 2)$ by complementation, and $(2, 1) \sim (1, 2)$ is configuration (II), where the same subtraction leaves a difference of single elements against a difference of pairs and the collapse does not bite. $\square$

The census backs the case analysis cell by cell. On the 267 non-empty pairs of the wide base: no twin with halved invariants, and no independent second twin — the environment uniqueness re-verified through the case analysis itself. Moreover, the cells closed by the linear argument alone are empty already at the linear level (as the proof demands), while the lemma-closed cells and the two residual cells host 2,512 *linear* candidates in all (882 in the transversal cell, 194 in the half-sum cell) — every one killed by the quadratic equation. The remaining cases are exactly configurations (I) and (II) — and the free census of the next subsection (up to $N = 26$) shows that *both are realized* on free bimagic pairs, with explicit members, and that in case (II) there are pairings exhibiting both $k = 3$ and $k = 2$ alignment: the two general conjectures are false. No exception occurs among the 1,078 column-block pairs surveyed; the case analysis identifies the residual configurations that Conjecture 16 asserts the square environment excludes.

Remark 19 (relation to the P.T.E. literature). *Twins are instances of the Prouhet–Tarry–Escott problem of degree 2 and size 3, for which solutions abound in general. The point of the uniqueness question is not the existence theory of P.T.E. pairs but the way two bimagic sextuples constrain the local P.T.E. structure: on free pairs the census finds the constraint stopping at $|G| \leq 4$, in two rigid forms (Section 7.5), and Conjecture 16 says that inside squares it is total. We are not aware of this question in the P.T.E. literature.*

7.5 The free census: both residual configurations are realized

To test Conjectures 11 and 14 outside the square environment we ran an exhaustive census of *free* bimagic pairs: all pairs of disjoint six-element subsets of $[1, N]$ with equal sum and equal sum of squares, for $N \leq 26$ (286,705 pairs at $N = 26$).

Twin counts. Throughout the census range $|G| \in \{0, 2, 4\}$: four twins occur, never more. As forced by complementation, every pair with $|G| = 2$ consists of a complementary pair (verified 11,158 of 11,158 at $N = 24$). Four-twin pairs number 156 in $[1, 22]$, 306 in $[1, 24]$, 564 in $[1, 26]$. Up to relabelling of the four twins, they fall into the two incidence forms identified by Proposition 18: (II) *transversal* (156, 300, 552 pairs), two complementary twin pairs meeting transversally, and (I) *half-sum* (0, 6, 12 pairs, none below $[1, 23]$), where both sextuples split into two halves with halved invariants and the four twins are the 2×2 pairings of the halves.

The exclusion property fails for the transversal form, whose realizations admit pairings with $N_3(\pi) > 0$ and $N_2(\pi) > 0$. Every transversal pair in the census shows the same alignment statistics over its 720 pairings: 32 mixed ($N_3 > 0$ and $N_2 > 0$), 40 purely row-aligned, 488 purely skew, 160 null. The smallest counterexample lies already in $[1, 15]$: $A = \{1, 3, 9, 10, 11, 14\}$, $B = \{2, 5, 6, 7, 13, 15\}$, whose order-preserving pairing has $N_3 = 2$ and $N_2 = 2$. Every half-sum pair in the census, by contrast, has alignment vector $(k, 3-k, 3-k, k)$ over its four twins: 72 pairings purely row-aligned, 648 purely skew, none mixed and none null — the exclusion holds for every pairing of a half-sum pair and fails for the transversal form. These statistics depend only on the incidence form, as shown by the abstract six-point

model built from the incidence matrix, which reproduces both tables by finite counting over all realizations of each form, with the row marginals 36/324/324/36 of Remark 17.

The generator of the half-sum form. Half-sum pairs arise from quadruples of pairwise disjoint triples with equal sum and equal sum of squares: each quadruple assembles into two sextuples in three ways, giving three four-twin pairs, and within the census *every* half-sum pair arises this way. Quadruples number 0 in $[1, 22]$, 1 in $[1, 23]$, then one more per step up to 4 in $[1, 26]$ — all four quadruples found in the census are translates of the single seed $\{1, 17, 18\}, \{2, 13, 21\}, \{3, 11, 22\}, \{6, 7, 23\}$, four triples sharing the invariants (36, 614).

The boundary. Within the free census range both residual configurations are realized and both general conjectures are refuted; on the 1,078 column-block pairs of the fifteen squares surveyed, $|G| \leq 2$ with no exception. What separates the surveyed square environment from the free census is therefore not the existence of the two residual forms — both occur in the census — but their absence from the surveyed column blocks (Conjecture 16).

8 Open problems

1. The global minimum of S_1 (and of S_2) over all semi-bimagic squares of order 6 with distinct integers, with MaxNb unconstrained. The trivial bound $S_1 \geq 111$ improves for free: $S_1 = 111$ forces the entry set $\{1, \dots, 36\}$, whose maximum $36 \leq 41$ is excluded by Theorem 2; hence $S_1 \geq 112$.
2. The analogous minimality question at order 7.
3. Prove Conjecture 16 (uniqueness in the square environment). By Theorem 13 this establishes the exclusion for every pairing throughout the environment. The two general conjectures are refuted by the free census (Section 7.5), which realizes both residual configurations of Proposition 18; the open question is what mechanism, if any, forces $|G| \leq 2$ in the square environment.

Remark 20 (What the data already rule out). *A natural probabilistic reading of the sieve — the four surviving sets are simply the richest in row partitions — fails. Mirror symmetry is empirically necessary for the optimal sets, but far from sufficient: 378 of the 382 mirror-symmetric candidates admitting row partitions yield no square. Exhaustive enumeration further shows that row-partition abundance alone cannot characterize survival: the global champion among all 45,985 candidates omits $\{1, 2, 3, 4, 15, 32\}$ (asymmetric, $S_1 = 141$) and admits 197 row partitions yet no square; 326 non-winners are richer than the poorest winner; and the four surviving sets admit between 50 and 132 row partitions, of which only 4–8 extend to a square. Restricted to the mirror-symmetric candidates, the winners rank 1, 8, 18, 24 by partition count. The unresolved problem is to identify the global compatibility invariant governing the intersection between row and column decompositions.*

Data availability

All enumeration code, raw logs (with the identifiers cited in the tables above), and the machine-readable list of the 324 partitions are available at [REPOSITORY]. For Section 7: `seduta-g.log` (twin census, realization check, exhaustive 32,400 sweep), `selezione-j3.log`

(the 1,200-pair dichotomy), `m5-predizioni.log` (pre-registered shuffle), `terza-biunivocita.log` and `terza-vaglio-unicita.log` (independent re-implementations of the bijection check and of the 1,078-pair uniqueness screen), and `residuo-ab.log` (the case-analysis census of Proposition 18). For the free census of Section 7.5: `fresco-vaglio.log` and `fresco-excl.log` (census over $[1, 22]$ and refutation sweep), `invariante-156.log`, `invariante-n24.log` and `salita-n26.log` (four-twin census to $N = 26$: incidence forms and alignment statistics), `halfsum-generatore.log` (the quadruple generator), `modello-tabella-fine.log` (the abstract-model alignment tables), and `cintura-finale.log` (the complementation check at $|G| = 2$).

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